Machine Learning: class error glossary

Start list, one row per error card. Students add the errors they meet.

You see	It means	Fix	Session
SyntaxError or invalid character	A quotation mark is curly, usually from copying out of a document.	Retype the quote marks with the keyboard so they are straight.	ML 01
ModuleNotFoundError: No module named 'sklearn'	This laptop is not using the prepared camp environment.	Tell the instructor. Do not install anything yourself.	ML 01
NameError: name 'price' is not defined	Python has not run a line that creates that name in this session.	Run the cell that creates the name first, and check the spelling and capital letters.	ML 02
TypeError: can't multiply sequence by non-int of type 'str'	You tried to multiply two pieces of text.	Remove the quotation marks around the number.	ML 02
ValueError: could not convert string to float	The text has letters, a comma or a currency symbol inside it.	Remove everything except digits and one decimal point.	ML 03
ZeroDivisionError: division by zero	The calculation divides by zero.	Check the number you divide by; it must not be 0.	ML 03
IndentationError: expected an indented block	The line after if, elif or else does not start with spaces.	Add four spaces at the start of each line inside the block.	ML 04
SyntaxError: invalid syntax	A colon is missing at the end of an if, elif or else line.	Put a colon at the end of the line.	ML 04
IndexError: list index out of range	You asked for a position the list does not have.	A list of 3 items has positions 0, 1 and 2.	ML 05
KeyError: 'price'	The dictionary has no key spelled like that.	Check the exact spelling and capital letters of the key.	ML 05
TypeError: total() missing 1 required positional argument	You gave the function fewer values than it has parameters.	Give one argument for each parameter.	ML 06
The answer shows None	The function has no return line, so nothing came back.	Add return before the value you want back.	ML 06
FileNotFoundError: No such file or directory	The CSV file is not beside the notebook.	Move the file into the same folder as the notebook, or ask the instructor.	ML 07
ModuleNotFoundError: No module named 'pandas'	The prepared camp environment is not active.	Tell the instructor. Do not install packages.	ML 07
KeyError: 'unit'	The column name is spelled differently from the header.	Print the headers with list(shop.columns) and copy the exact name.	ML 08
The filter returns an empty table	The comparison matches no rows, often because of a typo in a text value.	Check the spelling and capital letters of the value you compare with.	ML 08
NameError: name 'np' is not defined	numpy was not imported in this notebook session.	Run import numpy as np first.	ML 09

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The proportion is above 1	You divided the whole by the part.	Divide the part by the whole.	ML 09
FileNotFoundError when reading the log back	to_csv did not run, or ran in another folder.	Run the to_csv line again and check the file list in Jupyter.	ML 10
KeyError: 'price'	The column name has a different spelling in this table.	Print list(raw.columns) and copy the exact name.	ML 10
NameError: name 'sns' is not defined	seaborn was not imported in this session.	Run import seaborn as sns first.	ML 11
The chart does not appear	The cell that draws it has not run, or plt.show() is missing.	Run the import cell, then run the drawing cell again.	ML 11
The notebook works only until I restart it	A later cell uses a name that an earlier cell created and you deleted.	Restart the kernel and run all cells from the top to find the missing step.	ML 12
The totals do not agree	A filter or a typo dropped rows before grouping.	Compare the row counts before and after each step.	ML 12
NameError: name 'X' is not defined	The cell that creates X has not run yet.	Run the cells in order from the top.	ML 13
KeyError: 'area'	The column is called area_m2 in this file.	Print homes.columns and copy the exact name.	ML 13
ValueError: Expected 2D array, got 1D array instead	The model needs a table of inputs, with rows and columns, and got a flat list.	Use double brackets, such as [[35]], or select with homes[["area_m2"]].	ML 14
ModuleNotFoundError: No module named 'sklearn'	The prepared camp environment is not active.	Tell the instructor. Do not install packages.	ML 14
KeyError: "['rooms'] not in index"	The column name is spelled differently from the header.	Print homes.columns and copy the exact name.	ML 15
The two errors are the same	The model and baseline were scored on the same predictions, or the wrong table was used.	Check that each uses its own predict call on X_test.	ML 15
ValueError: could not convert string to float	A category name was passed where numbers are needed.	Use iris.target for the labels and iris.data for the inputs.	ML 16
NameError: name 'iris' is not defined	The cell that loads iris has not run.	Run the cells in order from the top.	ML 16
The two kinds of mistake sit in the wrong cells	The axes were read the wrong way round: rows are actual, columns are predicted.	Label both axes before reading any cell.	ML 17
The matrix has actual and predicted swapped	The two lists were passed to confusion_matrix in the wrong order. It expects the actual values first and the predicted values second.	Write confusion_matrix(actual, predicted) and check that the row totals match the real class counts.	ML 17
The final accuracy was calculated before the choice	The held-back set was used while deciding the depth.	Mark that result as validation and use the instructor's alternate holdout for the final.	ML 18

You see	It means	Fix	Session
ValueError: The least populated class in y has only 1 members	stratify was given labels that are too few.	Check that y is the full label list.	ML 18
KeyError: 'tip'	The column is called tip_ugx.	Print fares.columns and copy the exact names.	ML 19
The plan promises too much	It asks for real people's data, an app or a neural network.	Reduce it to one supplied table, one target and one baseline.	ML 19
Baseline and model score exactly the same	Both scores were computed from the same predictions, so the baseline is not a separate guess.	Build the baseline from the training mean alone and score the model with its own predict, both on X_val and y_val.	ML 20
ValueError: Found input variables with inconsistent numbers of samples	X and y have different lengths because one was filtered and not the other.	Create X and y from the same table at the same step.	ML 20
The with-tip score looks like the best	A leakage column made the model look better.	Remove the column that is only known after the answer.	ML 21
The comparison changes every time I run it	The split uses different random_state values or none at all.	Use the same split settings for every model.	ML 21
NameError after restarting the kernel	A cell used a name that was only created in a deleted or skipped cell.	Run all cells from the top and add the missing step.	ML 22
The final score differs from last time	The split or the model settings changed.	Use the same random_state and settings as in validation.	ML 22
The slides run over five minutes	There are more than five ideas.	Keep one idea per slide and cut the rest.	ML 23
The demonstration does not run	The notebook was not restarted or the data file was moved.	Show the screenshot backup and explain the steps.	ML 23
The change breaks the notebook	A name or a cell order changed.	Restart, run all cells, then read the last line of the error.	ML 24
The demonstration is too long	More than one idea per slide.	Use the five slides and leave the extra detail for questions.	ML 24